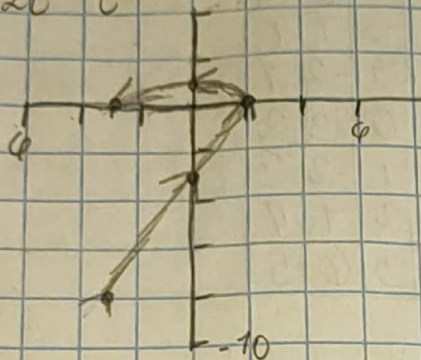


# Tarea Sección 10.1

1)  $X = 1 - t^2$   
 $Y = 2t - t^2$   
 $-1 \leq t \leq 2$



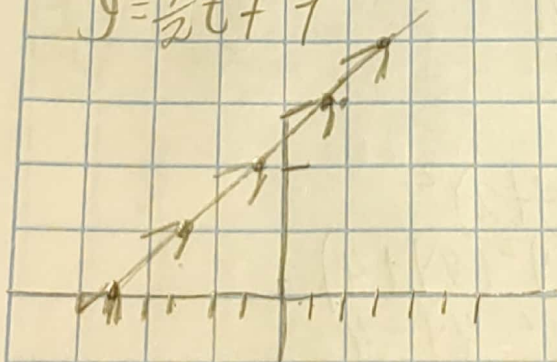
| t  | X  | Y  | $X = 1 - t^2$                    |
|----|----|----|----------------------------------|
| -2 | -3 | -8 | $X - 1 = -t^2$                   |
| -1 | 0  | -3 | $\sqrt{X-1} = t$                 |
| 0  | 1  | 0  |                                  |
| 1  | 0  | 1  | $2(\sqrt{X-1}) - (\sqrt{X-1})^2$ |
| 2  | -3 | 0  |                                  |

3)  $X = t + \sin t$   
 $Y = \cos t$   
 $-\pi \leq t \leq \pi$



| t        | X      | Y  |
|----------|--------|----|
| $-\pi$   | $-\pi$ | -1 |
| $-\pi/2$ | -2.5   | 0  |
| 0        | 0      | 1  |
| $\pi/2$  | 2.5    | 0  |
| $\pi$    | $\pi$  | -1 |

5)  $X = 2t - 1$   
 $Y = \frac{7}{2}t + 1$



| t  | X  | Y   |
|----|----|-----|
| -2 | -5 | 0   |
| -1 | -3 | 1/2 |
| 0  | -1 | 1   |
| 1  | 1  | 3/2 |
| 2  | 3  | 2   |

~~$Y = \frac{7}{2}t + 1$~~   
 ~~$4 - 1 = \frac{7}{2}t$~~   
 ~~$2(4 - 1) = t$~~

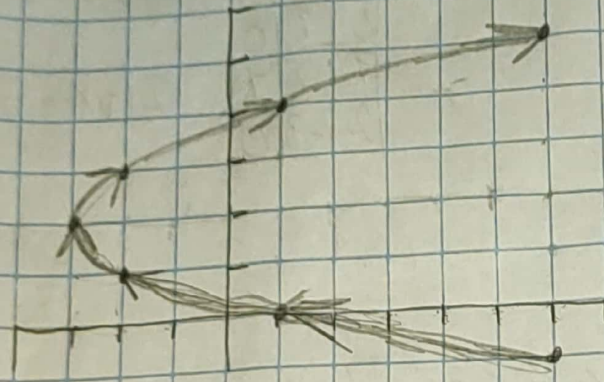
$X = 2t - 1$   
 $\frac{X + 1}{2} = t$

$Y = \frac{7}{2} \left( \frac{X+1}{2} \right) + 1$   
 $Y = \frac{7(X+1)}{4} + 1$

$X = 2$



7)  $x = t^2 - 3$   $-3 \leq t \leq 3$   
 $y = t + 2$



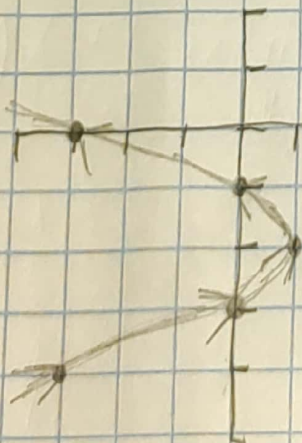
| t  | x  | y  |
|----|----|----|
| -3 | 6  | -1 |
| -2 | 1  | 0  |
| -1 | -2 | 1  |
| 0  | -3 | 2  |
| 1  | -2 | 3  |
| 2  | 1  | 4  |
| 3  | 6  | 5  |

$y = t + 2$   
 $y - 2 = t$

$x = t^2 - 3$   $y = \sqrt{x+3} + 2$   
 $\sqrt{x+3} = t$

$x = (y-2)^2 - 3$   
 $x = y^2 - 4y + 4 - 3$   
 $x = y^2 - 4y + 1$

9)  $x = 7 - t^2$   $-2 \leq t \leq 2$   
 $y = t - 2$



| t  | x  | y  |
|----|----|----|
| -2 | -3 | -4 |
| -1 | 0  | -3 |
| 0  | 7  | -2 |
| 1  | 0  | -1 |
| 2  | -3 | 0  |

$y = t - 2$   
 $y + 2 = t$

$x = 7 - (y+2)^2$   
 $x = 7 - (y^2 + 4y + 4)$   
 $x = 7 - y^2 - 4y - 4$   
 $x = -y^2 - 4y - 3$



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$$11) \begin{aligned} x &= \sin \frac{1}{2}\theta \\ y &= \cos \frac{1}{2}\theta \end{aligned} \quad -\pi \leq \theta \leq \pi$$

$$\sin^2 \frac{1}{2}\theta + \cos^2 \frac{1}{2}\theta = 1$$

$$x^2 + y^2 = 1$$



| $\theta$         | $x$                   | $y$                  |
|------------------|-----------------------|----------------------|
| $-\pi$           | $-1$                  | $0$                  |
| $-\frac{\pi}{2}$ | $-\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{2}}{2}$ |
| $0$              | $0$                   | $1$                  |
| $\frac{\pi}{2}$  | $\frac{\sqrt{2}}{2}$  | $\frac{\sqrt{2}}{2}$ |
| $\pi$            | $1$                   | $0$                  |

$$13) \begin{aligned} x &= 3 \cos t \\ y &= 3 \sin t \end{aligned} \quad 0 \leq t \leq \pi$$

$$\frac{x}{3} = \cos t$$

$$\frac{y}{3} = \sin t$$

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$



75)  $x = t^2$   
 $y = \ln t$

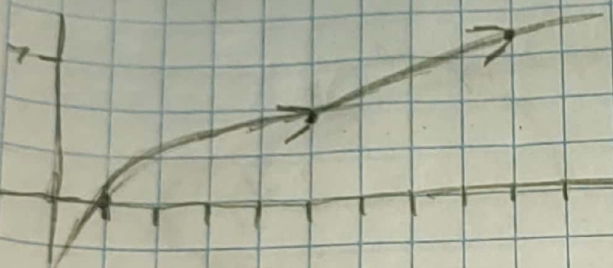
| t | x  | y    |
|---|----|------|
| 1 | 1  | 0    |
| 2 | 4  | 0.69 |
| 3 | 9  | 1.1  |
| 4 | 16 | 1.39 |

$$x = t^2$$

$$\sqrt{x} = t$$

$$y = \ln \sqrt{x}$$

$$y = \ln t$$



77)  $x = e^{2t}$   
 $y = t + 1$

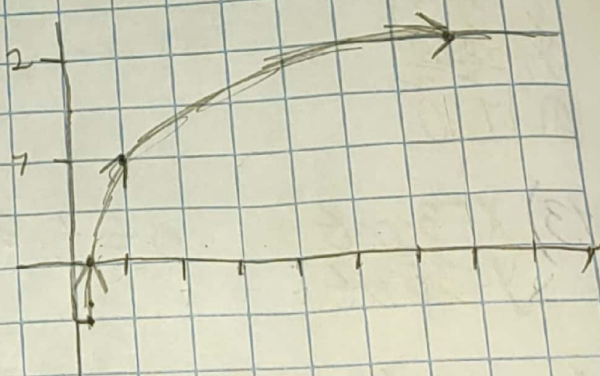
| t  | x     | y  |
|----|-------|----|
| -2 | 0.08  | -1 |
| -1 | 0.14  | 0  |
| 0  | 1     | 1  |
| 1  | 7.40  | 2  |
| 2  | 54.43 | 3  |

$$y - 1 = t$$

$$x = e^{2(y-1)}$$

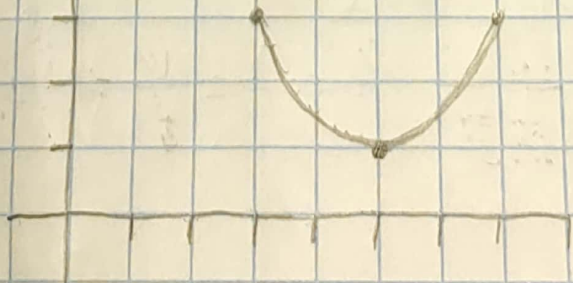
$$t = \frac{\ln |x|}{2}$$

$$y = \frac{\ln |x|}{2} + 1$$



79)  $x = 5 + 2 \cos \pi t$   $1 \leq t \leq 2$   
 $y = 3 + 2 \sin \pi t$

| t   | x | y |
|-----|---|---|
| 1   | 3 | 3 |
| 1.5 | 5 | 1 |
| 2   | 7 | 3 |



$$x - 5 = 2 \cos \pi t$$

$$\frac{x - 5}{2} = \cos \pi t$$

$$y - 3 = 2 \sin \pi t$$

$$\frac{y - 3}{2} = \sin \pi t$$

$$\cos^2 t + \sin^2 t = 1$$

$$\left(\frac{x-5}{2}\right)^2 + \left(\frac{y-3}{2}\right)^2 = 1$$

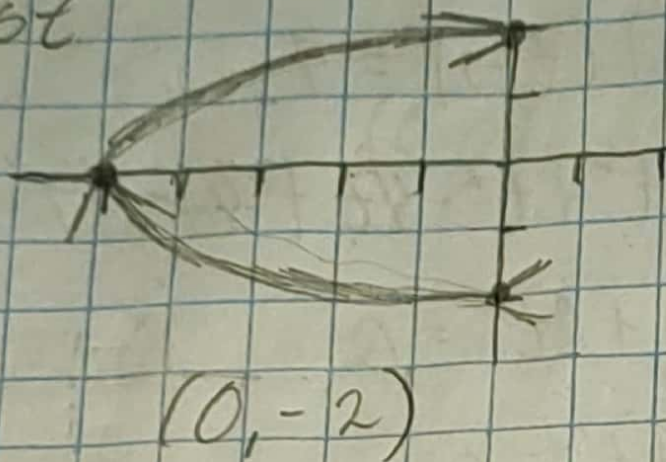
$$(3, 3) \quad (7, 3)$$

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21)  $x = 5 \sin t$   
 $y = 2 \cos t$

$$-\pi \leq t \leq 5\pi$$



$$\frac{x}{5} = \sin t \quad \frac{y}{2} = \cos t$$

$$\sin^2 t + \cos^2 t = 1$$

$$\left(\frac{x}{5}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

| t        | x  | y  |
|----------|----|----|
| $-\pi$   | 0  | -2 |
| $-\pi/2$ | -5 | 0  |
| 0        | 0  | 2  |
| $\pi$    | 0  | -2 |
| $2\pi$   | 0  | 2  |
| $3\pi$   | 0  | -2 |
| $4\pi$   | 0  | 2  |
| $5\pi$   | 0  | -2 |



$$23) \begin{cases} x = 5 \cos t \\ y = -5 \sin t \end{cases}$$

$$\frac{x^2}{25} + \frac{y^2}{25} = 1$$

$$dy/dx = 0$$

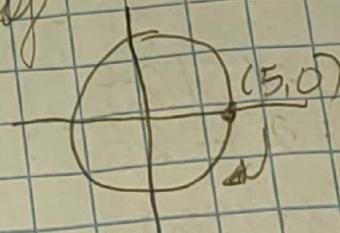
$$0 = -5 \sin t$$

$$0 = \sin t$$

$$t = 0 \quad t = 2\pi$$

$$t = \pi$$

$2\pi$  en el sentido  
de las agujas del  
reloj



$$25) \begin{cases} x = \sin^2 t \\ y = \cos^2 t \end{cases}$$

$$\sin^2 t + \cos^2 t = 1$$

$$x + y = 1$$



$$27) \begin{cases} x = \ln t \\ y = \sqrt{t} \end{cases}$$

$$t \geq 1$$

$$e^x = t$$

$$y = \sqrt{e^x}$$

$$y = e^{x/2}$$

